

BRENDAN JOYCE

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DATA ANALYST SUMMARY

Over 100 hours of graduate level data analysis coursework in Rstudio. Capable of generating reproducible research workflows. Comfortable taking raw data through cleaning, transformation, and visualization in tidyverse. Project work spans security and political analysis, including Janes Intelligence Review and Correlates of War Arms & Technology dataset.

TECHNICAL SKILLS

- **Languages & Tools:** tidyverse, dplyr, tidyr, ggplot2, lubridate, ggtext, knitr/kable, basic SQL.
- **Data Wrangling:** joins, pivots, regex cleaning, custom functions, HTML web scraping, rvest.
- **Visualization:** maps, bars, scatters, histograms, ribbon/area, patchwork, annotation, RevealJS.
- **Reproducible Research:** Quarto and Markdown, version control with Git/GitHub.

SELECTED DATA ANALYSIS PROJECTS

International Response to the 2022 Russian Invasion of Ukraine

- Used Janes Intelligence Review data to compare state-level political and diplomatic responses to the invasion.
- Joined and reshaped data, aggregated to state-level summaries.
- Generated a set of maps and short written analysis framed for a security-policy audience.

Military Modernization & Technology Diffusion (Armed UAVs)

- Built comparisons across weapon systems to trace how armed UAV capabilities have spread over time.
- Utilized coordinates from data to generate intuitive map visualizations.
- Combined multiple plots into a single figure with patchwork and wrote a short interpretation alongside it.

Webscrapping

- Wrote an R function to scrape U.S. senator profile data from C-SPAN website, parsing HTML elements.
- Used CSS selectors and regex to pull out keywords across the scraped text.

Replication & Improvement of Published Political Science Visualizations

- Replicated Figure 2 from 2021 Suppressing Black Votes case study on the Louisiana Understanding Clause.
- Redesigned axes, color scales, and annotations to make the original finding easier to read.
- Recreated a ribbon plot of Trump political popularity to practice plotting uncertainty bands.

Longitudinal Analysis of Global Environmental Change

- Reshaped the Greenspace dataset from raw observations into tidy tibbles.
- Developed plots to track land-cover change across time.

Custom Function

- Created a “compare to mean” function to show how each data point compares to the average.
- Allowed coders to quickly identify which values stand out from the average.

RELEVANT EDUCATION

American University, School of International Service

Master of Arts, International Affairs — Global Governance, Politics, and Security

Washington, DC

Expected May 2026

- **SIS 600 Intro to Statistics:** 75 hours learning basic statistical analysis techniques utilizing Rstudio.
- **SIS 750 Data Analysis:** 100 hours in Rstudio, wrangling, transformation, visualization, communication.